

PREFACE

This manual assumes that the MC68881/MC68882 is connected as a coprocessor to the MC68020/MC68030 microprocessor. If the MC68881/MC68882 is used in a system with a main processor other than the MC68020/MC68030, it is expected that the main processor emulates the M68000 Family coprocessor interface as required by the MC68881/MC68882.

This manual is divided into two major parts. The first part, sections 2 through 8, describes the programmer's model of the MC68881/MC68882 and the floating-point instruction set that it implements. This part of the manual includes a detailed description of each instruction and a section on instruction timing that can be used for program optimization and to predict floating-point arithmetic performance.

The second part of the manual, sections 9 through 13, describes the hardware interface of the MC68881/MC68882 to the main processor, and is most pertinent to system hardware designers. Bus cycle timing diagrams, interface register addressing, etc., are discussed from the viewpoint of the MC68020/MC68030 hardware conventions. A prior knowledge of the MC68020/MC68030 bus interface, particularly as it pertains to the M68000 Family coprocessor interface, is quite helpful in understanding the operation of the MC68881/MC68882 bus interface.

Throughout this manual, M68000 or M68000 Family is used to refer to the family of devices that support the Freescale 68000 Family architecture. A number that is preceded by MC, such as MC68020, MC68030, MC68881, or MC68882, refers to a specific part. A reference to MC68881/MC68882 or FPCP applies to either floating-point coprocessor, and a reference to MC68020/MC68030 or MPU applies to either main processor.

The sections and appendices of the manual are:

- Section 1. General Description
- Section 2. Programming Model
- Section 3. Operand Data Formats
- Section 4. Instruction Set
- Section 5. Coprocessor Programming
- Section 6. Exception Processing
- Section 7. Coprocessor Interface
- Section 8. Instruction Execution Timing
- Section 9. Functional Signal Descriptions
- Section 10. Bus Operation
- Section 11. Interfacing Methods
- Section 12. Electrical Specifications
- Section 13. Ordering Information and Mechanical Data
- Appendix A. Glossary
- Appendix B. Abbreviations and Acronyms

